

DINWIDDIE, VA, USA

WMO: 724014

Lat: 37.183N

Lon: 77.500W

Elev: 193

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	19.2	-0.3	5.4	25.6	5.0	7.2	29.3	20.2	47.1	18.3	46.6	2.3	350	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	97.2	76.8	93.4	75.9	91.2	74.9	80.5	91.1	79.0	89.8	77.8	87.7	6.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.8	145.3	85.3	75.5	134.4	83.4	74.9	131.8	82.7	44.1	91.3	42.6	90.0	41.1	88.0	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.7	15.4	12.7	DB	6.3	100.5	6.1	3.7	1.9	103.1	-1.7	105.3	-5.1	107.3	-9.5	110.0
			WB	5.7	82.3	6.0	2.5	1.3	84.1	-2.2	85.6	-5.6	87.0	-10.0	88.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	39.0	41.6	48.8	59.0	67.3	75.2	79.2	77.5	71.1	59.7	49.2	42.1
	DBStd	16.41	10.90	9.66	9.86	8.98	7.55	6.07	4.84	4.89	6.29	8.47	8.96	9.34
	HDD50	1226	373	268	145	19	0	0	0	0	15	123	283	
	HDD65	3692	808	657	511	225	68	5	0	0	209	479	711	
	CDD50	4600	31	32	108	290	536	757	907	854	633	315	98	39
	CDD65	1590	1	1	8	46	139	312	442	389	202	44	4	2
	CDH74	16840	5	8	113	617	1520	3324	4879	3996	1862	471	38	7
	CDH80	7485	1	0	19	197	565	1513	2456	1896	698	138	2	0
(14) Wind	WSAvg	4.7	5.3	5.6	6.0	6.1	4.8	4.0	3.5	3.4	4.2	4.2	4.6	4.6
(15) Precipitation	PrecAvg	45.3	3.4	2.7	4.3	3.4	4.0	3.7	4.5	4.6	4.5	3.3	3.2	3.7
	PrecMax	68.9	7.3	6.3	8.0	7.0	10.7	7.9	7.9	13.9	13.9	7.1	9.4	9.0
	PrecMin	34.2	0.9	0.8	0.2	1.3	1.1	0.9	1.3	1.3	0.4	0.0	0.2	0.7
	PrecStd	8.1	1.5	1.3	1.8	1.5	2.0	1.8	1.8	2.5	3.4	1.9	2.2	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.0	74.4	82.3	90.3	93.3	99.3	101.5	99.4	93.4	90.0	79.0	73.5
		MCWB	62.7	61.3	63.3	67.1	72.9	76.3	76.8	78.7	74.6	72.4	64.6	64.7
	2%	DB	67.1	70.1	77.1	85.3	90.0	94.9	97.5	95.7	90.4	83.5	73.3	68.5
		MCWB	60.0	57.2	60.1	65.6	72.3	76.3	77.5	76.8	73.8	70.0	62.0	62.4
	5%	DB	63.2	64.1	72.6	81.5	86.3	91.2	93.7	92.6	87.9	79.3	70.3	63.5
		MCWB	56.7	54.3	58.4	64.0	71.1	75.2	76.7	75.7	73.0	67.4	61.7	57.1
	10%	DB	57.1	59.5	67.3	76.8	82.3	88.8	91.0	90.0	83.9	74.8	65.9	58.8
		MCWB	50.6	50.3	56.3	61.6	68.7	74.2	75.8	74.8	71.6	65.1	59.2	52.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	64.6	67.0	72.5	77.1	80.5	82.9	82.2	79.4	75.7	70.2	67.7
		MCDB	69.9	70.9	76.0	85.5	88.5	91.9	93.6	93.2	88.0	82.6	73.8	70.6
	2%	WB	62.0	60.6	63.9	69.1	74.8	78.8	80.9	80.2	77.1	72.7	66.3	63.3
		MCDB	66.0	66.8	71.4	79.6	84.9	90.4	91.2	90.9	84.4	79.4	70.8	66.8
	5%	WB	57.6	56.4	61.5	66.7	72.7	77.6	79.3	78.5	75.4	70.3	63.4	58.9
		MCDB	62.6	62.8	70.9	76.3	82.6	88.1	90.3	88.5	82.5	77.2	67.0	63.1
	10%	WB	51.6	51.4	57.9	64.1	71.0	76.0	77.9	77.1	73.8	67.3	59.8	53.5
		MCDB	56.7	58.9	65.7	72.9	80.2	85.1	88.1	86.2	80.5	73.2	65.4	57.0
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	23.1	24.5	25.9	24.1	22.7	22.3	22.1	22.4	24.7	24.6	21.6
		MCDBR	27.4	30.8	30.5	30.0	28.0	26.3	25.7	25.4	26.4	28.5	29.2	26.3
	5% WB	MCWBR	20.2	20.7	16.9	13.9	13.4	11.5	10.1	10.1	11.7	15.8	18.5	19.6
		MCWBR	24.9	27.1	25.3	25.8	24.0	22.9	22.1	22.0	20.5	23.0	22.3	22.4
(40) Clear-Sky Solar Irradiance	taub		0.299	0.313	0.345	0.388	0.428	0.485	0.506	0.507	0.404	0.354	0.319	0.303
		taud	2.535	2.489	2.435	2.328	2.241	2.128	2.090	2.071	2.350	2.485	2.537	2.559
	Ebn at Noon		285	293	292	283	272	256	249	246	268	274	274	276
		Edh at Noon	24	29	34	40	44	49	51	51	36	29	24	23
(44) All-Sky Solar Radiation	RadAvg		749	985	1299	1641	1854	2009	1910	1728	1409	1111	839	658
	RadStd		61	112	105	129	162	135	95	87	140	135	66	57

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (73 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+268	+175	

Nomenclature: See separate page